IGCSE 0607 Practice Questions — E5.3

Similarity

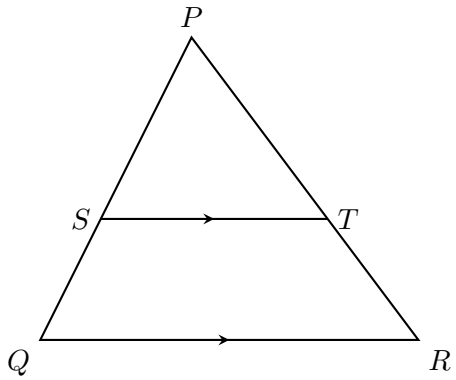
Total Questions: 39 — **Total Marks:** 264 Marks

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- **E5.3** Similarity
  → **Paper 2 (Non-Calculator)** *(16 Qs)* · **Paper 4 (Calculator)** *(23 Qs)*

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1 The diagram shows triangle PQR.
Point S lies on PQ and point T lies on PR.
The line ST is parallel to the line QR.

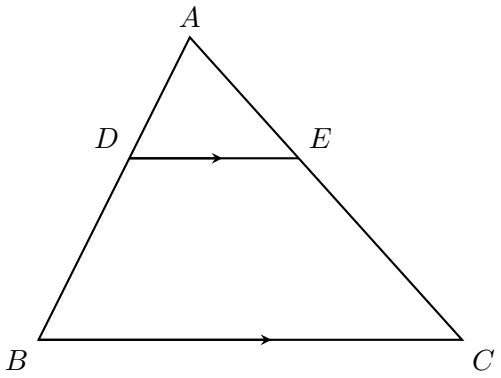


NOT TO SCALE

Given that $PS = 6$ cm, $SQ = 3$ cm and $ST = 8$ cm, calculate the length of QR.

$QR =$ ..... cm **[2]**

2 The diagram shows triangle ABC with a line DE drawn parallel to BC.
$AD = 4$ cm, $DB = 6$ cm, and $DE = 5$ cm.



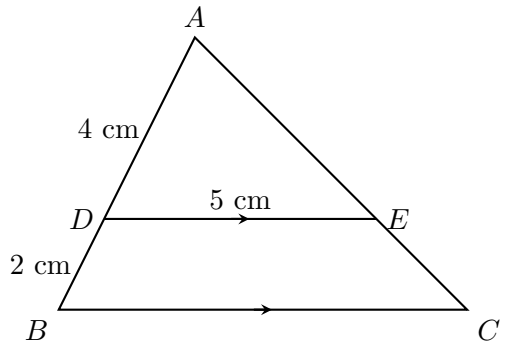
(a) Calculate the length of BC.

$BC =$ ..... cm **[2]**

(b) The area of triangle ADE is 16 cm^2.
Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

3 The diagram shows triangle ABC.
D is a point on AB and E is a point on AC such that DE is parallel to BC.
$AD = 4$ cm, $DB = 2$ cm and $DE = 5$ cm.



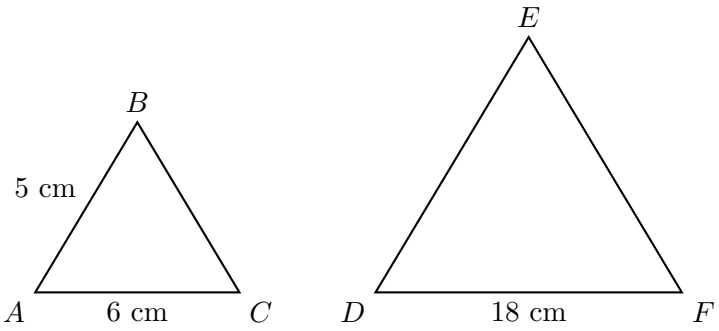
(a) Calculate the length of BC.

..... *cm* **[2]**

(b) The area of triangle ADE is 8 cm^2.
Work out the area of the trapezium $DBCE$.

..... *cm*^{2} **[2]**

4 Triangle ABC and triangle DEF are mathematically similar.



NOT TO SCALE

(a) Calculate the length DE.

$DE =$ ..... cm **[2]**

(b) The area of triangle DEF is 72 cm^2.
Calculate the area of triangle ABC.

..... cm^2 **[2]**

5 Two mathematically similar solid cylinders are shown as Cylinder A and Cylinder B.
Cylinder A has a height of 6 cm.
Cylinder B has a height of 15 cm.

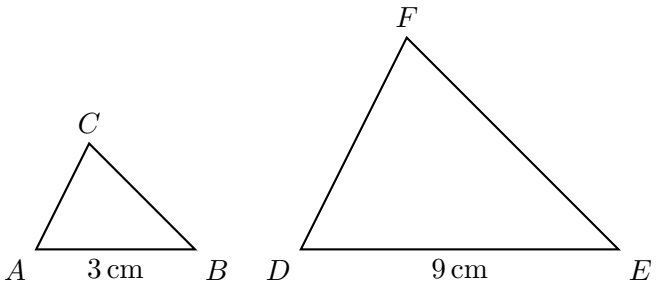
(a) The volume of Cylinder A is 48 cm^3.
Calculate the volume of Cylinder B.

..... cm^3 **[2]**

(b) The curved surface area of Cylinder B is 200 cm^2.
Calculate the curved surface area of Cylinder A.

..... cm^2 **[2]**

6 Triangle ABC and triangle DEF are mathematically similar.
$AB = 3$ cm and $DE = 9$ cm.



NOT TO SCALE

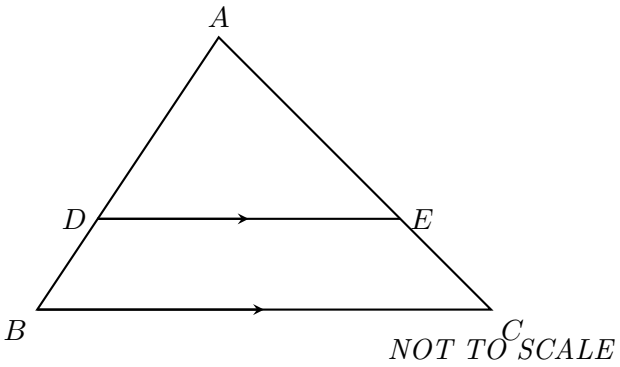
(a) The length of BC is 5 cm.
Calculate the length of EF.

..... cm **[2]**

(b) The area of triangle DEF is 54 cm^2.
Calculate the area of triangle ABC.

..... cm^2 **[2]**

7 In the diagram, ADE is a triangle and DE is parallel to BC.
$AD = 4$ cm, $DB = 2$ cm and $DE = 5$ cm.



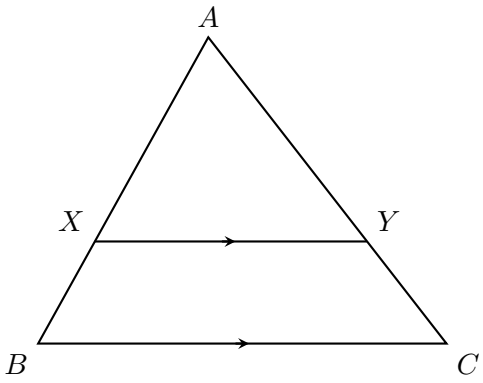
(a) Calculate the length of BC.

$BC =$ ..... cm **[2]**

(b) The area of triangle ADE is 8 cm^2.
Calculate the area of the trapezium $DBCE$.

Area = ... cm^2 **[3]**

8 In the diagram, ABC is a triangle and the line XY is parallel to BC.
$AX = 4$ cm, $XB = 2$ cm and $XY = 6$ cm.



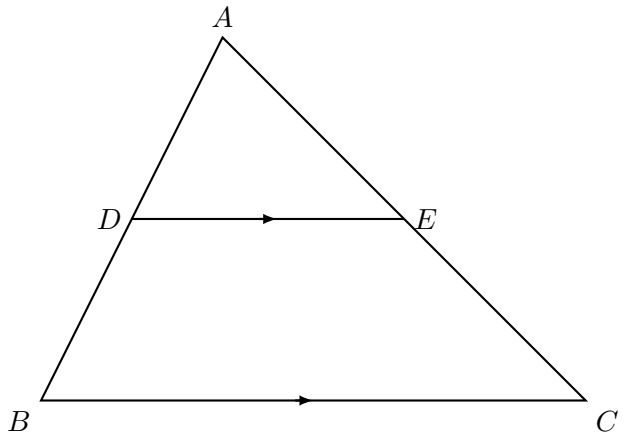
(a) Calculate the length of BC.

$BC =$ ..... cm **[2]**

(b) The area of triangle AXY is 32 cm^2.
Calculate the area of the trapezium $XBCY$.

Area = ... cm^2 **[3]**

9 The diagram shows a triangle ABC.
D is a point on AB and E is a point on AC such that DE is parallel to BC.
$AD = 4$ cm, $DB = 2$ cm and $DE = 6$ cm.



(a) Calculate the length of BC.

..... cm **[2]**

(b) The area of triangle ADE is 12 cm^2.
Calculate the area of the trapezium $DBCE$.

..... cm^2 **[3]**

10 Two mathematically similar bottles have heights of 10 cm and 15 cm.

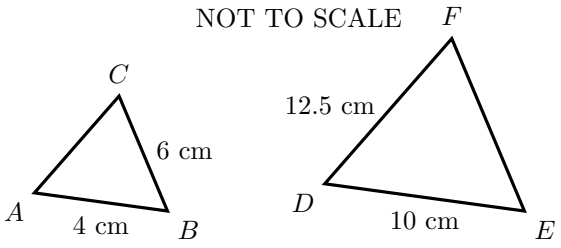
(a) The surface area of the smaller bottle is 120 cm^2.
Work out the surface area of the larger bottle.

..... cm^2 **[3]**

(b) The volume of the larger bottle is 540 cm^3.
Work out the volume of the smaller bottle.

..... cm^3 **[3]**

11 The diagram shows two mathematically similar triangles, ABC and DEF.
$AB = 4$ cm, $BC = 6$ cm, $DE = 10$ cm, and $DF = 12.5$ cm.



(a) Calculate the length of EF.

$EF =$ ..... cm **[2]**

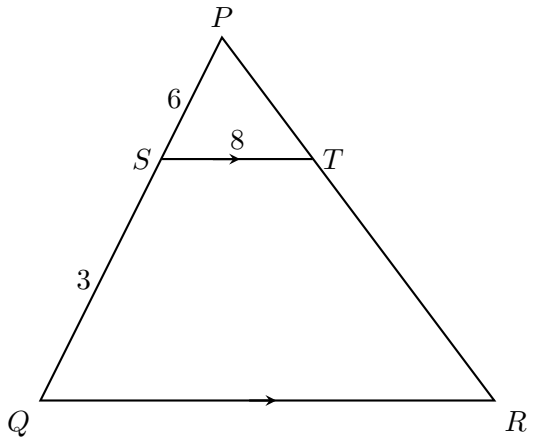
(b) Calculate the length of AC.

$AC =$ ..... cm **[2]**

(c) Given that the area of triangle ABC is 10 cm^2, calculate the area of triangle DEF.

..... cm^2 **[2]**

12 In the diagram, the line ST is parallel to QR.
$PS = 6$ cm, $SQ = 3$ cm and $ST = 8$ cm.
The length of PR is 12 cm.



(a) Calculate the length of QR.

..... cm **[2]**

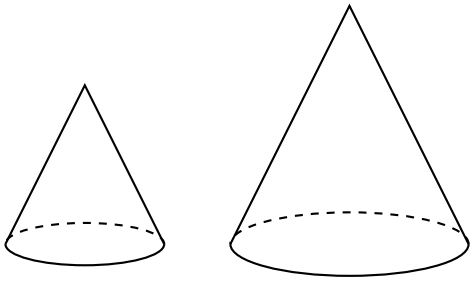
(b) Calculate the length of PT.

..... cm **[2]**

(c) The area of triangle PST is 20 cm^2.
Calculate the area of the trapezium $SQRT$.

..... cm^2 **[3]**

13 Cone A and Cone B are mathematically similar.
The total surface area of Cone A is 20 cm^2 and the total surface area of Cone B is 45 cm^2.



Cone A Cone B

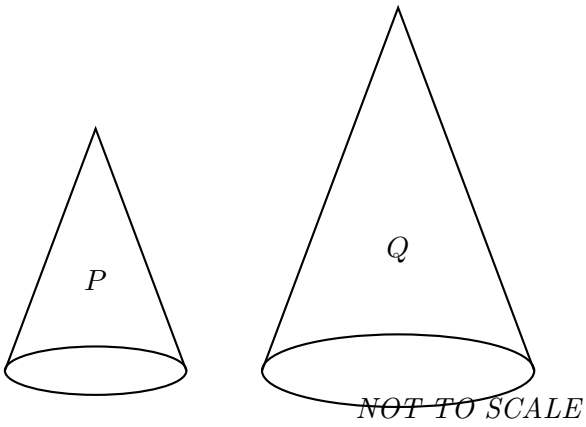
(a) The volume of Cone A is 16 cm^3.
Calculate the volume of Cone B.

..... cm^3 **[3]**

(b) The height of Cone B is 12 cm.
Calculate the height of Cone A.

..... cm **[2]**

14 The diagram shows two similar cones, P and Q.
Cone P has a volume of 400 cm^3 and a height of 12 cm.
Cone Q has a volume of 1350 cm^3.



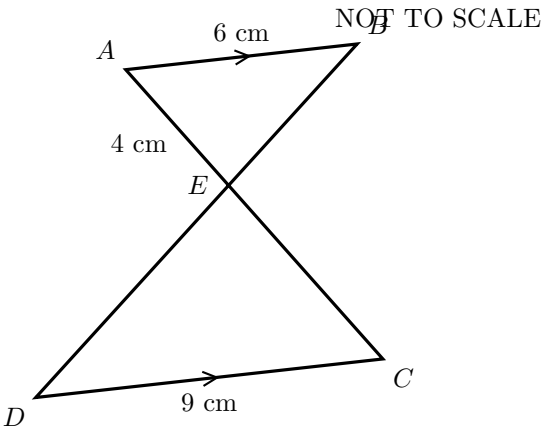
(a) Calculate the height of cone Q.

Height = ... cm **[3]**

(b) The curved surface area of cone Q is 360 cm^2.
Calculate the curved surface area of cone P.

Surface area = ... cm^2 **[2]**

15 In the diagram, the straight lines AC and BD intersect at E, and AB is parallel to DC.
$AB = 6$ cm, $DC = 9$ cm, $AE = 4$ cm, and $BD = 12.5$ cm.



(a) Explain why triangle ABE is similar to triangle CDE.

.....

.....

..... **[3]**

(b) Calculate the length of

(i) EC,

$EC =$ ..... cm **[2]**

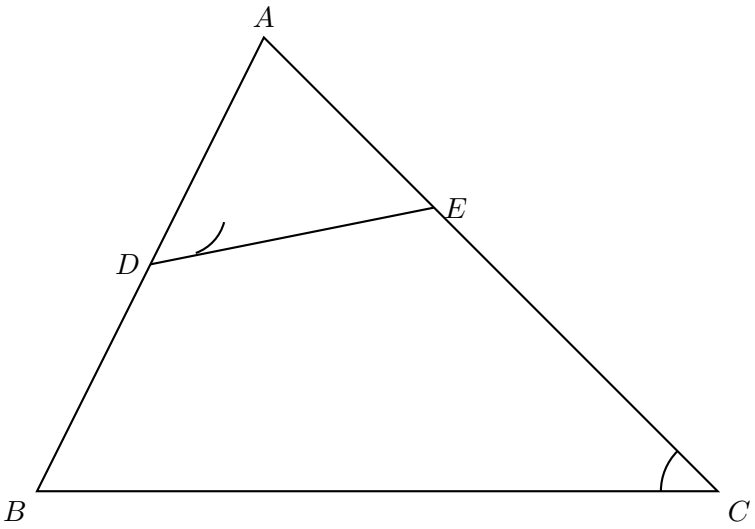
(ii) BE.

$BE =$ ..... cm **[2]**

(c) Given that the area of triangle ABE is 12 cm^2, calculate the area of triangle CDE.

..... cm^2 **[2]**

16 The diagram shows a triangle ABC.
The point D lies on AB and the point E lies on AC.
The angle ADE is equal to the angle ACB.



(a) Explain why triangle ADE is mathematically similar to triangle ACB.

[3]

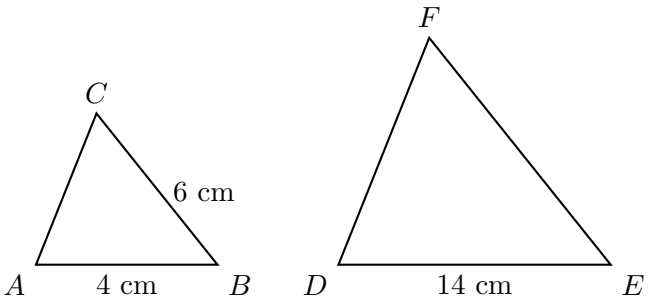
(b) It is given that $AD = x$ cm, $DB = (x + 1)$ cm, $AE = 4$ cm and $EC = 5$ cm.
Find the value of x.

$x =$ ..... **[4]**

(c) Find the ratio of the area of triangle ADE to the area of the quadrilateral $DBCE$.
Give your answer in its simplest form.

..... **[2]**

17 Triangles ABC and DEF are mathematically similar.
In the diagram, $AB = 4$ cm, $BC = 6$ cm and $DE = 14$ cm.



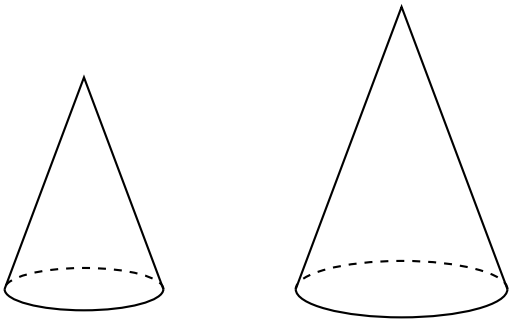
(a) Calculate the length of EF.

..... cm **[2]**

(b) The area of triangle ABC is 8.5 cm^2.
Calculate the area of triangle DEF.

..... cm^2 **[2]**

18 Cone A and Cone B are mathematically similar.
The surface area of Cone A is 450 cm^2 and the surface area of Cone B is 800 cm^2.



Cone A Cone B

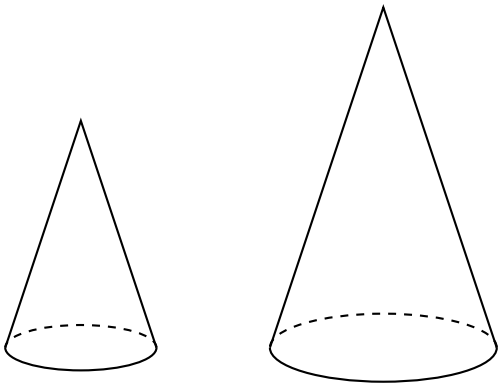
(a) The volume of Cone A is 1350 cm^3.
Calculate the volume of Cone B.

Volume = ... cm^3 **[3]**

(b) The height of Cone B is 48 cm.
Calculate the height of Cone A.

Height = ... cm **[2]**

19 Two solid cones, A and B, are mathematically similar.



Cone A Cone B

(a) Cone A has a volume of 150 cm^3 and a height of 12 cm.
Cone B has a volume of 1200 cm^3.
Calculate the height of cone B.

..... cm **[3]**

(b) The surface area of cone B is 440 cm^2.
Calculate the surface area of cone A.

..... cm^2 **[2]**

20 Two mathematically similar containers, A and B, have volumes of 320 cm^3 and 1080 cm^3 respectively.

(a) Container A has a height of 18 cm.

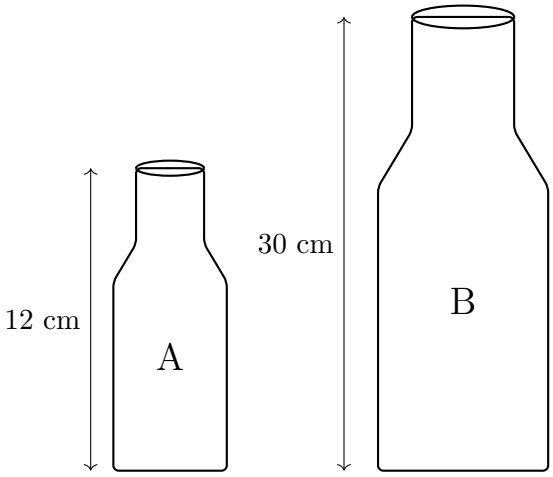
Calculate the height of container B.

Height = ... cm **[3]**

(b) Container B has a surface area of 630 cm^2.
Calculate the surface area of container A.

Surface area = ... cm^2 **[3]**

21 Two mathematically similar containers, A and B, are shown in the diagram.
Container A has a height of 12 cm and container B has a height of 30 cm.



NOT TO SCALE

(a) Container A has a volume of 320 cm^3.
Calculate the volume of container B.

Volume = ... cm^3 **[2]**

(b) Container B has a surface area of 1125 cm^2.
Calculate the surface area of container A.

Surface area = ... cm^2 **[2]**

(c) The containers are completely filled with liquid.
The cost of filling container A is \$4.80.
Assuming the cost is directly proportional to the volume, calculate the cost of filling container B.

..... **[2]**

22 Two mathematically similar solids, P and Q, have volumes of 320 cm^3 and 8640 cm^3 respectively.

(a) The height of solid P is 12 cm.
Calculate the height of solid Q.

..... cm **[3]**

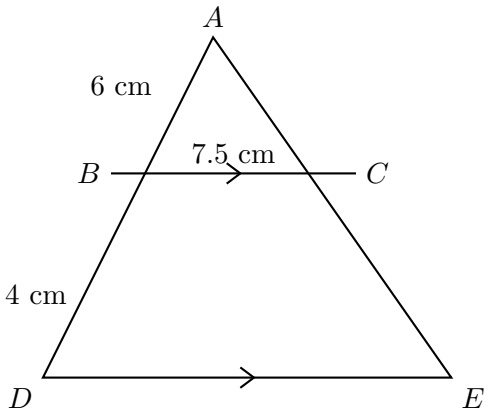
(b) The surface area of solid Q is 2250 cm^2.
Calculate the surface area of solid P.

..... cm^2 **[2]**

(c) Solid P is made of a metal with a density of 8.5 g/cm^3.
Calculate the mass of solid P in kilograms.

..... kg **[2]**

23 The diagram shows triangle ADE with a line BC drawn parallel to DE.
$AB = 6$ cm, $BD = 4$ cm and $BC = 7.5$ cm.



NOT TO SCALE

(a) Explain why triangle ABC is mathematically similar to triangle ADE.

[2]

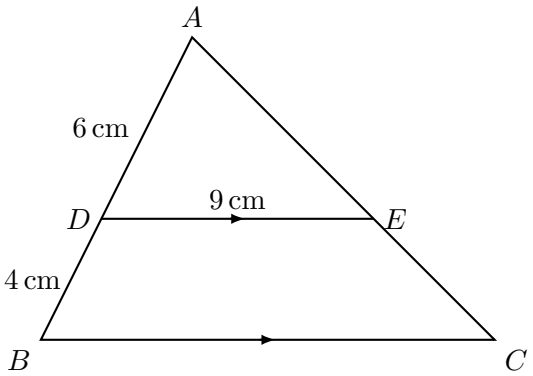
(b) Calculate the length of DE.

$DE =$ ..... cm **[2]**

(c) The area of triangle ABC is 18 cm^2.
Calculate the area of the quadrilateral $BCED$.

..... cm^2 **[3]**

24 In the diagram, DE is parallel to BC.
$AD = 6$ cm, $DB = 4$ cm and $DE = 9$ cm.



NOT TO SCALE

(a) Explain why triangle ADE is similar to triangle ABC.

.....

.....

..... **[2]**

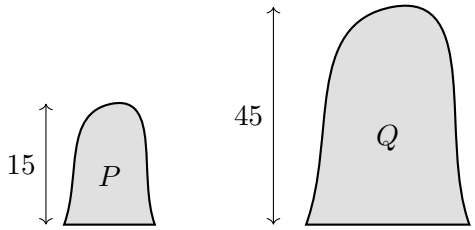
(b) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(c) The area of triangle ADE is 27 cm^2.
Calculate the area of the trapezium $DBCE$.

..... cm^2 **[3]**

25 The diagram shows two mathematically similar solid statues, P and Q.
Statue P has a height of 15 cm and a mass of 500 g.
Statue Q has a height of 45 cm.
Assume that both statues are made from the same uniform material.



(a) Calculate the mass of statue Q.
Give your answer in kilograms.

..... kg **[3]**

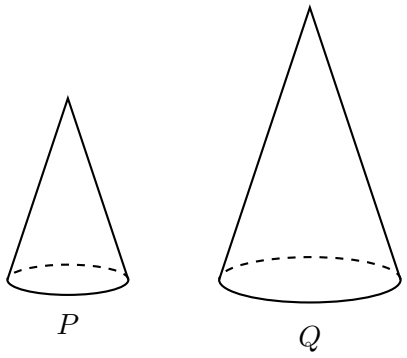
(b) A third similar statue, R, has a mass of 4 kg.
Calculate the height of statue R.

..... cm **[3]**

(c) The surface area of statue Q is 2430 cm^2.
Calculate the surface area of statue P.

..... cm^2 **[2]**

26 The diagram shows two geometrically similar solid metal cones, P and Q.
The volume of cone P is 320 cm^3 and the volume of cone Q is 1080 cm^3.



(a) The height of cone P is 12 cm.
Calculate the height of cone Q.

..... cm **[3]**

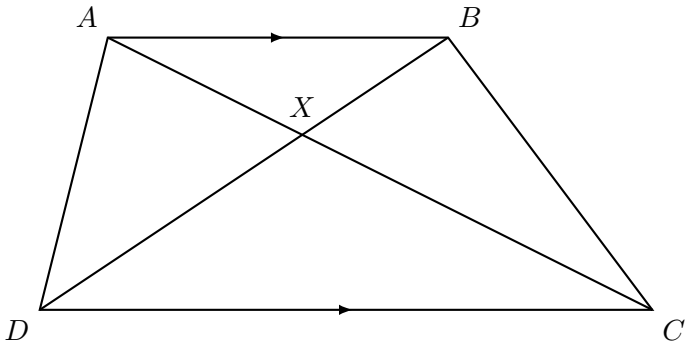
(b) The surface area of cone Q is 450 cm^2.
Calculate the surface area of cone P.

..... cm^2 **[2]**

(c) Both cones are melted down to make a single new solid cone R, which is geometrically similar to P and Q.
Calculate the height of cone R.
Give your answer correct to 3 significant figures.

..... cm **[3]**

27 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The diagonals AC and BD intersect at point X.
$AB = 9$ cm, $DC = 15$ cm and $BX = 6$ cm.
The area of triangle ABX is 27 cm^2.



NOT TO SCALE

(a) Explain why triangle ABX is similar to triangle CDX.

[2]

(b) Calculate the length of BD.

..... cm **[3]**

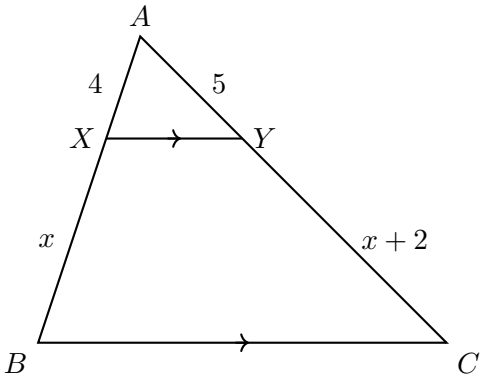
(c) Calculate the area of triangle CDX.

..... cm^2 **[2]**

(d) Calculate the area of triangle BCX.

..... cm^2 **[2]**

28 In the diagram, the line XY is parallel to BC.
$AX = 4$ cm, $XB = x$ cm, $AY = 5$ cm and $YC = (x + 2)$ cm.



(a) Calculate the value of x.

$x =$ ..... **[3]**

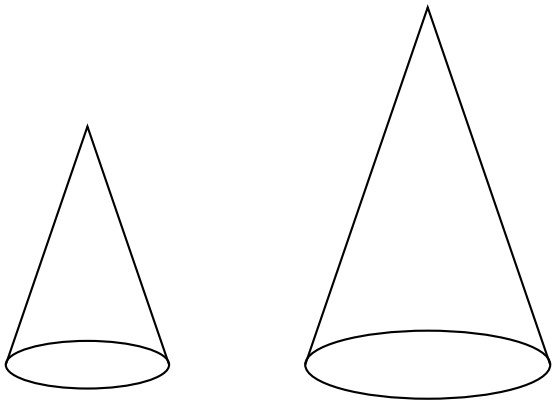
(b) $XY = 3.5$ cm.
Calculate the length of BC.

..... cm **[2]**

(c) The area of triangle AXY is A cm^2.
Find an expression for the area of the trapezium $XBCY$ in terms of A.
Give your answer in its simplest form.

..... **[2]**

29 Cone A and Cone B are mathematically similar solids.



NOT TO SCALE

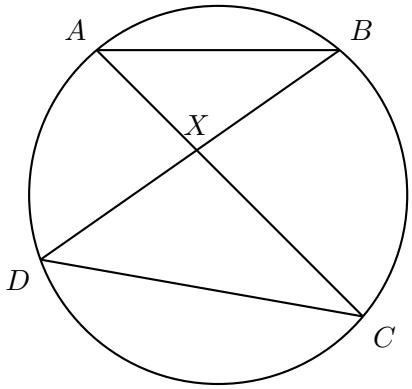
(a) The volume of Cone A is 108 cm^3 and the volume of Cone B is 500 cm^3.
The sum of the surface areas of Cone A and Cone B is 340 cm^2.
Calculate the surface area of Cone B.

..... cm^2 **[4]**

(b) Cone B has a total vertical height of 20 cm.
A smaller cone, mathematically similar to Cone B, is removed from the top of Cone B to leave a frustum.
The volume of the frustum is $\frac{7}{8}$ of the volume of Cone B.
Calculate the vertical height of the removed small cone.

..... cm **[3]**

30 The diagram shows a circle with points A, B, C and D on its circumference.
The chords AC and BD intersect at X.



(a) Show that triangle AXB is similar to triangle DXC.
Give a geometric reason for each statement you make.

[3]

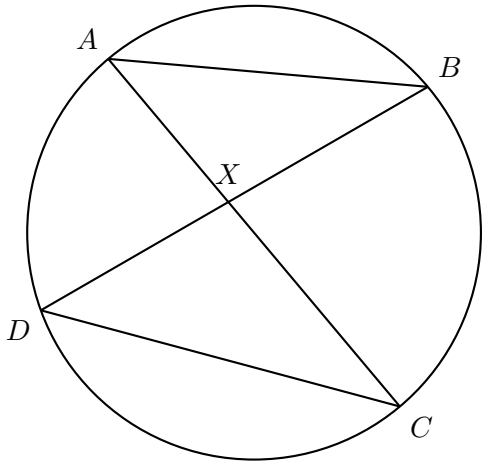
(b) $AX = 4$ cm, $BX = 5$ cm, $CX = 7.5$ cm and $AB = 6$ cm.
Calculate the length of DX and the length of DC.

$DX =$ ..... , $DC =$ ..... **[3]**

(c) The area of triangle AXB is 11.8 cm^2.
Calculate the area of triangle DXC.

..... cm^2 **[2]**

31 Points A, B, C, and D lie on the circumference of a circle.
The straight lines AC and BD intersect at the point X.



(a) Show that triangle AXB is similar to triangle DXC, giving geometrical reasons for your statements.

[3]

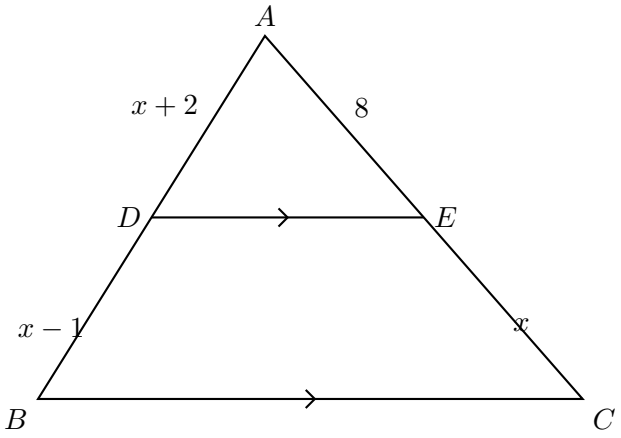
(b) $AX = (x + 2)$ cm, $CX = (x - 2)$ cm, $BX = (x - 3)$ cm, and $DX = (x + 4)$ cm.
Calculate the value of x.

$x =$ ..... **[3]**

(c) The area of triangle DXC is 45 cm^2.
Calculate the area of triangle AXB.

Area = ... cm^2 **[2]**

32 In the diagram, DE is parallel to BC.
$AD = (x + 2)$ cm, $DB = (x - 1)$ cm, $AE = 8$ cm and $EC = x$ cm.



NOT TO SCALE

(a) Show that $x^2 - 6x + 8 = 0$.

[3]

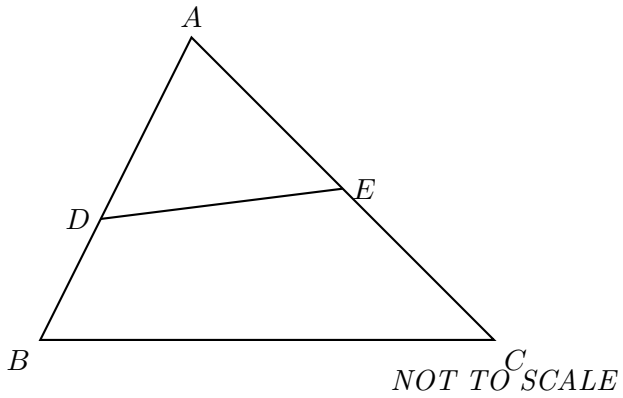
(b) Solve the equation $x^2 - 6x + 8 = 0$.

$x =$ ..... , $x =$ ..... **[2]**

(c) Given that the area of triangle ADE is 24 cm^2 and $x = 4$, calculate the area of the trapezium $DBCE$.

..... cm^2 **[3]**

33 In the diagram, D is a point on AB and E is a point on AC.
Angle AED = Angle ABC.
$AD = 6$ cm, $AE = 5$ cm and $DB = 9$ cm.



(a) Show that triangle ADE is similar to triangle ACB.
Give reasons for your statements.

[3]

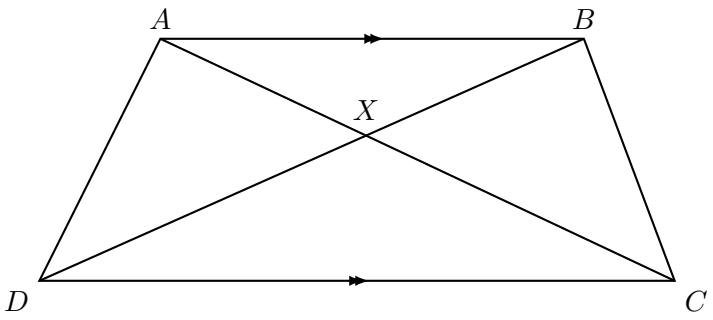
(b) Calculate the length of EC.

$EC =$ ..... cm **[3]**

(c) Find the ratio of the area of triangle ADE to the area of quadrilateral $BCED$ in its simplest form.

..... **[2]**

34 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The diagonals AC and BD intersect at X.
$AB = 7$ cm and $DC = 10.5$ cm.



(a) Explain why triangle ABX is similar to triangle CDX.

[2]

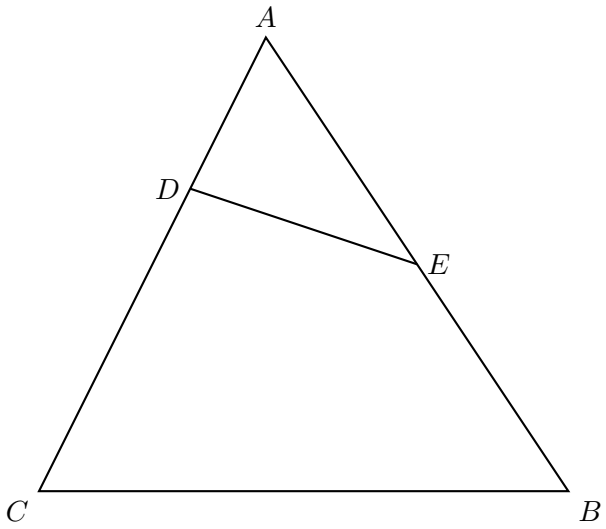
(b) $AX = 4$ cm.
Calculate the length of AC.

$AC =$ ..... cm **[3]**

(c) The area of triangle ABX is 16 cm^2.
Calculate the area of the trapezium $ABCD$.

..... cm^2 **[4]**

35 The diagram shows triangle ABC.
D is a point on AC and E is a point on AB.
Angle ABC = Angle AED.



NOT TO SCALE

(a) Explain why triangle ABC is similar to triangle AED.

.....

.....

..... **[2]**

(b) Given that $AC = 15$ cm and $AD = 6$ cm, write down the scale factor of the enlargement that maps triangle AED onto triangle ABC.

..... **[1]**

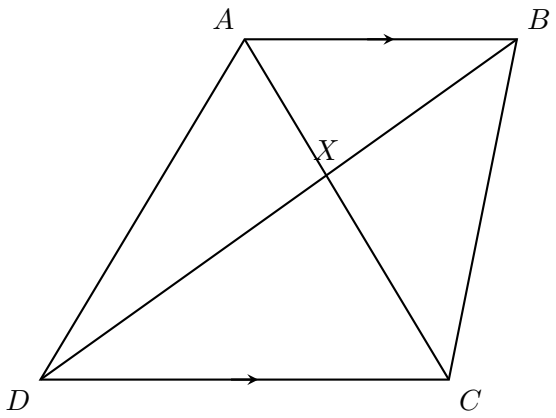
(c) $AE = x$ cm and $AB = (2x + 2.5)$ cm.
Use similarity to set up an equation and find the value of x.

$x =$ ..... **[3]**

(d) The area of the quadrilateral $BCDE$ is 42 cm^2.
Calculate the area of triangle ADE.

..... cm^2 **[3]**

36 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The diagonals AC and BD intersect at X.



(a) Explain, giving geometric reasons, why triangle ABX is similar to triangle CDX.

[3]

(b) $AX = 4$ cm and $CX = 6$ cm.
$AB = x$ cm and $DC = (2x - 3)$ cm.

Work out the value of x.

$x =$ ..... **[3]**

(c) The area of triangle ABX is 24 cm^2.
Calculate the area of the trapezium $ABCD$.

Area = ... cm^2 **[4]**

37 Two solid statues, A and B, are mathematically similar.

(a) Statue A has a height of 12 cm and a volume of 800 cm^3.
Statue B has a volume of 2700 cm^3.
Calculate the height of Statue B.

..... cm **[3]**

(b) The cost to paint a statue is directly proportional to its surface area.
It costs \$18 to paint Statue A.
Calculate the cost to paint Statue B.

\$ ..... **[3]**

(c) A third mathematically similar statue, C, has a surface area that is 44% greater than the surface area of Statue A.
Calculate the volume of Statue C.

..... cm^3 **[4]**

38 Solid Cone A has a base radius of 10 cm and a perpendicular height of 24 cm.

(a) Calculate the total surface area of Cone A.
Give your answer in terms of π.

..... cm^2 **[3]**

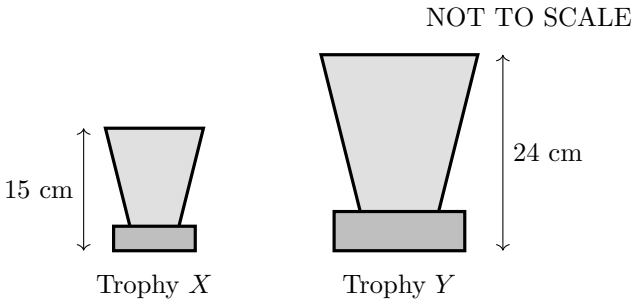
(b) Cone B is mathematically similar to Cone A.
The volume of Cone B is 100π cm^3.
Calculate the total surface area of Cone B.
Give your answer in terms of π.

..... cm^2 **[4]**

(c) A frustum is made by cutting a smaller cone, which is geometrically identical to Cone B, from the top of Cone A.
This frustum is melted down and recast completely into a solid sphere.
Calculate the radius of this sphere.

..... cm **[4]**

39 The diagram shows two mathematically similar solid metal trophies, Trophy X and Trophy Y.
Trophy X has a height of 15 cm and a total surface area of 320 cm^2.
Trophy Y has a height of 24 cm and a volume of 2048 cm^3.



(a) Calculate the total surface area of Trophy Y.

..... cm^2 **[2]**

(b) Calculate the volume of Trophy X.

..... cm^3 **[3]**

(c) A third trophy, Trophy Z, is mathematically similar to Trophy X and has a volume of 1372 cm^3.
Calculate the height of Trophy Z.

..... cm **[3]**

(d) Trophy Y is melted down to make as many complete copies of Trophy X as possible.
The remaining metal is used to make a solid sphere of radius r cm. Calculate the value of r, giving your answer correct to 3 significant figures.

$r =$ ..... **[4]**